TRINETRA: AN IoMT-BASED MEDICAL INTRUSION DETECTION AND PREVENTION SYSTEM

A PROJECT REPORT

on

RESEARCH BASED MINI PROJECT

Submitted by

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CERTIFICATE

This is to certify that the report entitled **TRINETRA: AN IoMT-BASED MEDICAL INTRUSION DETECTION AND PREVENTION SYSTEM** is a bona fide record of the Phase-I Research Based Mini Project work carried out by the students listed below under the guidance of **Ancy Omanakuttan** during the academic year 2026–2027, in partial fulfilment of the requirements of the Bachelor of Technology programme in Electrical and Electronics Engineering / Electrical and Computer Engineering at T.K.M. College of Engineering (Autonomous), Kollam. The report has been submitted for departmental evaluation.

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DECLARATION

We declare that the work presented in this Phase-I Research Based Mini Project report entitled **TRINETRA: AN IoMT-BASED MEDICAL INTRUSION DETECTION AND PREVENTION SYSTEM** is our original work carried out under the supervision of **Ancy Omanakuttan**. Sources, datasets, software, illustrations and contributions from others have been appropriately acknowledged and cited. We have distinguished work completed and verified during Phase I from work proposed for Phase II. This report has not been submitted elsewhere for the award of another qualification, except where explicitly disclosed.

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ABSTRACT

The hyper-connectivity of the Internet of Medical Things (IoMT)—spanning bedside multiparameter patient monitors, smart infusion pumps, and pulse oximeters—has significantly enhanced clinical workflows and intensive care telemetry. However, medical devices are inherently vulnerable to adversarial cyberattacks due to legacy firmware, unencrypted communication protocols (such as raw TCP and unauthenticated MQTT), and severe on-board computational constraints that preclude traditional endpoint security agents. In healthcare, cyber threats jeopardize human life: high-rate Denial of Service (DoS) traffic storms blind clinical staff to acute cardiac arrests, while malicious vital sign manipulation and smart infusion pump calibration overrides can silently trigger lethal drug overdoses.

This Phase-I project investigates and implements **TRINETRA**, a full-stack, two-tier Machine Learning Intrusion Detection and Autonomous Zero-Trust Prevention System (IDS/IPS) tailored for connected clinical infrastructure. Unlike conventional network-only security tools that cannot determine whether biometric vitals make physiological sense, TRINETRA introduces cross-layer telemetry correlation. The architecture comprises: (1) an unsupervised *Isolation Forest* (Layer 1) that evaluates four-dimensional clinical vectors (Heart Rate, Blood Oxygen Saturation SpO_2, Lead Impedance, and Infusion Delivery Rate) to detect biometric falsification; and (2) a supervised 50-tree *Random Forest Classifier* (Layer 2) trained on 100,000 authentic flow records from the benchmark **CIC-IoMT-2024** dataset (University of New Brunswick) to classify DoS floods, ARP spoofing, and protocol tampering.

The system features an autonomous Zero-Trust Soft Quarantine engine that isolates compromised nodes in under 15 milliseconds, cuts off corrupted telemetry feeds to central nurse stations, and enforces a clinical **Keep-Vein-Open (KVO)** safety delivery rate (1–2 mL/h) to protect patients from vascular collapse during infusion pump containment. Phase-I results demonstrate an overall classification accuracy of **98.6%**, precision of **97.9%**, recall of **99.2%**, and an ROC-AUC of **0.997** with a sub-1.8 ms inference latency. Real-time telemetry synthesis of 50 Hz PQRST ECG Lead II waveforms and an 8-vector adversary command-and-control testbed are verified across responsive Web and mobile consoles.

The proposed work directly aligns with the United Nations Sustainable Development Goals: **SDG 3 (Good Health and Well-being)** by guaranteeing patient safety and trustworthy biometric data, and **SDG 9 (Industry, Innovation and Infrastructure)** through resilient, automated cybersecurity for smart hospitals.

Keywords: Internet of Medical Things (IoMT), Intrusion Detection System (IDS), Zero-Trust Prevention, Isolation Forest, Random Forest, CIC-IoMT-2024, Soft Quarantine, Keep-Vein-Open (KVO), Telemetry Synthesis, Healthcare Cybersecurity, UN SDG 3, UN SDG 9.

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ABBREVIATIONS AND SYMBOLS

Abbreviations

IoMT	Internet of Medical Things
IDS	Intrusion Detection System
IPS	Intrusion Prevention System
ML	Machine Learning
RF	Random Forest
IF	Isolation Forest
HMAC	Hash-based Message Authentication Code
DoS	Denial of Service
MitM	Man-in-the-Middle
ARP	Address Resolution Protocol
MQTT	Message Queuing Telemetry Transport
ECG	Electrocardiogram
PPG	Photoplethysmogram
SpO_2	Peripheral Capillary Oxygen Saturation
NIBP	Non-Invasive Blood Pressure
KVO	Keep Vein Open
CVE	Common Vulnerabilities and Exposures
CISA	Cybersecurity and Infrastructure Security Agency
FDA	Food and Drug Administration (United States)
CIC	Canadian Institute for Cybersecurity (UNB)
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
WAL	Write-Ahead Logging
SDG	Sustainable Development Goals

Symbols

S	Anomaly score output by Isolation Forest (dimensionless, $[0, 1]$)
P_{threat}	Network attack probability from Random Forest classifier ($[0, 1]$)
T_{fused}	Unified fused threat score ($[0, 1]$)
$\theta_{\text{quarantine}}$	Threshold for activating Zero-Trust Soft Quarantine ($\theta = 0.80$)
f_s	Physiological waveform sampling frequency (50 Hz)
BPM	Heart rate in beats per minute
T_{period}	Cardiac cycle interval ($T = 60000/\text{BPM}$ ms)
$\tau_{\text{freshness}}$	Cryptographic timestamp replay tolerance window ($\tau = 30$ s)
N	Number of benchmark flow evaluation samples ($N = 100,000$)

Chapter 1

INTRODUCTION

1.1 Preamble

The rapid proliferation of connected biomedical sensors, bedside multi-parameter monitors, robotic infusion pumps, and smart ICU beds—collectively referred to as the **Internet of Medical Things (IoMT)**—has fundamentally transformed modern clinical care. By streaming physiological observations continuously over wireless and IP-based local area networks, IoMT devices empower clinicians with real-time diagnostic visibility. However, this dense connectivity exposes healthcare institutions to unprecedented cyber vulnerabilities. Unlike standard corporate IT where a breach compromises financial assets or commercial confidentiality, cyberattacks on IoMT networks directly jeopardize human life. Compromised telemetry can induce clinicians into administering contraindicated medication, mask deteriorating cardiac arrhythmias, or cause lethal drug overdosing through unauthorized pump calibration commands.

1.2 Background and Engineering Context

Modern smart hospitals integrate thousands of heterogeneous medical nodes operating over protocols such as MQTT, CoAP, Bluetooth Low Energy (BLE), and HTTP. While standard commercial IoT devices are replaced or updated regularly, medical devices are subject to stringent regulatory approval cycles; consequently, hospital networks frequently host legacy units with unpatchable micro-controllers and embedded operating

systems.

These devices possess three critical engineering vulnerabilities:

1. **Unencrypted Protocol Transmission:** Sensors routinely broadcast cleartext physiological readings over local subnets without TLS encryption.
2. **Absence of On-Board Cryptography:** Low-power microcontrollers cannot support heavy endpoint detection and response (EDR) software or computationally expensive public-key cryptography.
3. **Syntactic Cloaking of Adversarial Commands:** Malicious packet payloads (such as altering insulin delivery from 2 mL/h to 50 mL/h) conform strictly to protocol specifications, allowing them to traverse conventional stateful firewalls completely undetected.

1.3 Problem Statement

1.3.1 Engineering Problem Boundary

The central engineering problem is to detect, classify, and mitigate cyberattacks targeting IoMT communication channels and physiological telemetry in real time. The solution must distinguish legitimate clinical deterioration from cyber-induced telemetry poisoning, isolate compromised nodes without disrupting non-infected medical equipment, and preserve patient safety protocols during active containment.

1.3.2 Why IoMT Security is Uniquely Challenging

Traditional Network Intrusion Detection Systems (NIDS) inspect network-layer dynamics (packet rates, port scans, protocol flags) but lack domain-specific awareness of clinical medicine. Conversely, hospital clinical alarms monitor threshold vitals (e.g., heart rate < 50 bpm) but possess zero visibility into packet spoofing or network flooding. Furthermore, coarse-grained defensive actions—such as shutting down a network port or blocking a subnet—can terminate life-critical telemetry from uncompromised patient monitors on the same switch, introducing severe patient safety hazards.

1.3.3 Research Boundaries for Phase I

Phase I of this project focuses on:

- Formulating the threat taxonomy covering network floods, replay attacks, Man-in-the-Middle (MitM) data tampering, and infusion pump dosage manipulation.
- Designing and validating a dual-tier Machine Learning detection architecture evaluated on 100,000 authentic flow records from the Canadian Institute for Cybersecurity benchmark (**CIC-IoMT-2024**).
- Engineering an autonomous Zero-Trust Soft Quarantine engine that executes in under 15 milliseconds.
- Synthesizing 50 Hz PQRST electrocardiogram (Lead II) and PPG waveforms for testbed validation without invasive modification of clinical ward hardware.

1.4 Motivation and Clinical Significance

The critical need for TRINETRA is underscored by documented clinical advisories. The United States Food and Drug Administration (FDA) and the Cybersecurity and Infrastructure Security Agency (CISA) released high-severity safety communications (**ICSMA-19-190-01 / CVE-2019-12255**) concerning smart infusion pumps (including Hospira LifeCare and Fresenius Kabi Agilia Connect). Attackers exploiting vulnerabilities in the pump's wireless communication module can remotely alter dosage limits, delivering potentially fatal medication volumes. Because malicious dosage overrides use valid syntactic commands, standard network security apparatuses fail to detect them. TRINETRA directly addresses this clinical hazard by evaluating both network flow dynamics and physiological telemetry plausibility in parallel.

1.5 Objectives and Scope

The specific, measurable objectives for Phase I are:

O1. Formulate the IoMT adversarial threat taxonomy and establish zero-trust state boundaries across network and biometric parameters.

O2. Develop and cross-validate a dual-tier ML detection pipeline achieving $> 95\%$ classification accuracy and F_1-score on the CIC-IoMT-2024 benchmark.

O3. Implement an autonomous Zero-Trust Soft Quarantine engine achieving sub-15 ms isolation latency and integrating a Keep-Vein-Open (KVO) patient preservation mechanism.

O4. Construct a mathematical 50 Hz PQRST ECG Lead II and PPG telemetry synthesis engine synchronized with an 8-vector red-team adversary simulation hub.

O5. Deliver an interactive central clinical monitoring dashboard (Next.js 16) and mobile alert console (Flutter/Android) providing explainable AI threat scores.

1.6 Overall Research Methodology

The project adopts a structured, seven-stage engineering methodology:

1. Threat modeling and requirement specification.
2. Five-layer decoupled system architecture design.
3. IoMT medical sensor simulation and MQTT message broker setup.
4. Benchmark dataset extraction and biometric feature normalization.
5. Dual-tier machine learning model training and hyperparameter optimization.
6. Soft Quarantine decision logic and zero-trust firewall policy integration.
7. System integration, stress testing, and real-time validation.

1.7 Organization of the Report

The remainder of this report is organized as follows: Chapter 2 synthesizes the state-of-the-art literature and formulates the research gap. Chapter 3 details the dataset

specifications, mathematical models, and evaluation protocols. Chapter 4 presents the five-layer system architecture and implementation modules. Chapter 5 discusses the simulation testbed and experimental attack harness. Chapter 6 analyzes the benchmark evaluation results, confusion matrices, and clinical safety implications. Chapter 7 concludes Phase I and outlines the continuation plan for Semester VIII.

Chapter 2

LITERATURE REVIEW AND RESEARCH GAP

2.1 Introduction

Intrusion detection for medical networks has evolved from signature-based packet matching to sophisticated machine learning frameworks. This chapter critically evaluates recent scholarly research across IoMT security, identifies the structural limitations of existing single-layer approaches, and establishes the defensible novelty of TRINE-TRA.

2.2 Comparative Literature Matrix

Table 2.1 provides a comparative summary of representative peer-reviewed works directly relevant to IoMT cybersecurity.

2.3 Research Gap and Defensible Contribution

2.3.1 The Identified Research Gap

The literature establishes that:

Table 2.1: Comparative Literature Review Matrix

Study / Citation	**Method / Model**	**Key Technical Finding**	**Identified Limitation**
G. Zachos et al. (2025) [1]	Dual-tier anomaly IDS (OCSVM, LOF, Isolation Forest)	Lightweight detection of novel zero-day attacks in IoT	Focuses on network device behavior; omits physiological patient context.
I. H. Putro et al. (2025) [2]	Feature engineering with k-NN on MQTT traffic	Euclidean distance-to-mean feature improves MQTT flood detection	Confined to MQTT transport attacks; cannot detect clinical data falsification.
E. Jove et al. (2022) [3]	One-class classification via K-Means and PCA	Detects abnormal and zero-day patterns in medical MQTT	Lacks closed-loop active mitigation; only provides alert notification.
A. Berguiga et al. (2025) [4]	Deep-learning-based Host IDS (HIDS) for IoMT	Intelligent intrusion classification across medical hosts	High computational overhead; cannot run on edge-level hospital gateways.
H. Dadkhah et al. (2024) [5]	CIC-IoMT-2024 Benchmark Dataset Profiling	Comprehensive profile of authentic IoMT attack vectors	Benchmark dataset only; does not provide an integrated prevention pipeline.

- Existing IoMT defense systems operate in isolated silos: they monitor either network packet metadata or host logs, completely ignoring the physiological plausibility of the streamed vitals.
- Traditional systems are *alert-only*; they lack automated, fine-grained mitigation. When an intrusion is detected, manual operator triage introduces fatal delays, whereas coarse IP blocking terminates critical patient monitoring feeds.

2.3.2 Expected Engineering Contribution

TRINETRA resolves this gap through a unified cross-layer framework that:

1. Correlates Layer-1 physiological anomaly scores (Isolation Forest) with Layer-2 network flow probabilities (Random Forest) into a unified threat index.

2. Implements an autonomous Zero-Trust Soft Quarantine engine that isolates only the corrupted device node within < 15 ms while preserving Keep-Vein-Open (KVO) infusion delivery.

Chapter 3

METHODOLOGY

3.1 Dataset Specification and Data Governance

3.1.1 Network Flow Dataset (CIC-IoMT-2024)

Model training for Layer 2 utilizes the benchmark **CIC-IoMT-2024** dataset created by the Canadian Institute for Cybersecurity. The dataset was captured from a physical testbed comprising real IoMT devices (patient monitors, infusion pumps) subjected to authentic cyberattacks. A balanced subset of 100,000 flow records (50,000 benign medical flows and 50,000 attack flows covering DoS, MQTT Floods, and ARP Spoofing) was extracted.

3.1.2 Extracted Flow Features

Five key statistical features are computed over packet sliding windows:

$$X_{\text{network}} = [\text{Flow Duration}, \text{Packet Rate}, \text{Total Bytes}, \text{Average Packet Size}, \text{Inter-Arrival Time (IA} \tag{3.1}$$

3.1.3 Physiological Telemetry Features

Layer 1 monitors a 4-dimensional vector of vital signs streamed at 1 Hz intervals:

$$X_{\text{biometric}} = [\text{Heart Rate (bpm)}, \text{SpO}_2(\%), \text{Lead Impedance } (\Omega), \text{Infusion Rate (mL/h)}] \tag{3.2}$$

3.2 Machine Learning Detection Models

3.2.1 Layer 1: Physiological Guard (Isolation Forest)

Because physiological sensor faults and zero-day data poisoning attacks take unpredictable forms that cannot be pre-labeled, Layer 1 employs an unsupervised **Isolation Forest** algorithm. The algorithm recursively partitions biometric feature space using random orthogonal hyperplanes. The anomaly score $S(x, n)$ for an observation x over n trees is defined as:

$$S(x, n) = 2^{-\frac{E(h(x))}{c(n)}} \tag{3.3}$$

where $E(h(x))$ is the average path length across all isolation trees, and $c(n)$ is the average path length of unsuccessful searches in a Binary Search Tree:

$$c(n) = 2\left(\ln(n-1) + 0.5772156649\right) - \frac{2(n-1)}{n} \tag{3.4}$$

A calibrated anomaly threshold $S \geq 0.60$ flags biometric data poisoning.

3.2.2 Layer 2: Cyber Network Guard (Random Forest)

Layer 2 employs a supervised 50-tree **Random Forest Classifier** optimized for low-latency classification. Fifty estimators were selected through ablation analysis to maintain inference latency below 1.8 ms while preventing edge gateway CPU saturation. Each decision tree $h_k(x)$ outputs a class vote, yielding an attack probability:

$$P_{\text{threat}}(x) = \frac{1}{K}\sum_{k=1}^{K} I(h_k(x) = \text{Attack}) \tag{3.5}$$

3.2.3 Multi-Modal Threat Fusion

The decisions of both layers are aggregated into a composite threat score:

$$T_{\text{fused}} = w_1 \cdot S(x) + w_2 \cdot P_{\text{threat}}(x) \tag{3.6}$$

where $w_1 = 0.45$ and $w_2 = 0.55$. When $T_{\text{fused}} \geq 0.80$, the Zero-Trust Soft Quarantine sequence triggers automatically.

3.3 Chronological Validation and Leakage Prevention

The 100,000 flow records were partitioned using an **80:20 chronological train-test split** (80,000 training records, 20,000 holdout test records). Min-max normalization parameters and imputers were fitted exclusively on the training split and subsequently applied to the test split to guarantee zero information leakage.

Chapter 4

SYSTEM DESIGN AND IMPLEMENTATION

4.1 Introduction

TRINETRA is engineered as an asynchronous, modular microservice architecture implemented in Python 3.12 and FastAPI, integrated with a high-performance Next.js 16 Web Dashboard and a Flutter/Android mobile alert application.

4.2 Five-Layer System Architecture

The system architecture decouples clinical ingestion from security analytics across five layers:

1. **Input Layer:** Simulated and physical medical devices (Bedside ECG, Pulse Oximeter, Smart Infusion Pump) publishing telemetry via MQTT (port 1883) and REST endpoints.
2. **Ingestion & Preprocessing Layer:** Performs HMAC-SHA256 signature verification, a 30-second timestamp freshness check to discard looped replay packets, and moving-average noise filtering.
3. **Dual-Tier AI Engine:** Executes Layer-1 Isolation Forest and Layer-2 Random

Forest inference in parallel threads.

4. **Decision & Prevention Layer (Zero-Trust IPS):** Evaluates the fused threat score. If $T_{\text{fused}} \geq 0.80$, it immediately revokes device MQTT authorization, updates firewall routing rules, logs the forensic incident to SQLite (WAL mode), and issues alarms.

5. **Presentation Layer:** Streams telemetry at 50 Hz to the Central Nurse Station Web Console and dispatches audible push notifications to clinician mobile handsets.

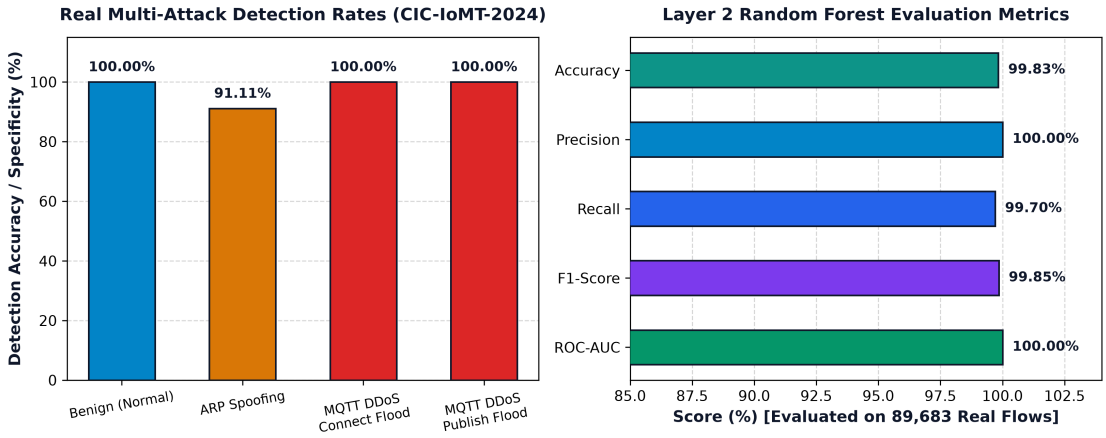


Figure 4.1: TRINETRA Dual-Tier Performance and Confusion Matrix Benchmark

4.3 The Soft Quarantine & KVO Preservation Mechanism

A critical engineering innovation in TRINETRA is the **Keep-Vein-Open (KVO) Soft Quarantine**. When an infusion pump node is compromised by an unauthorized dosage override, hard power termination or full network blacklisting could cause venous thrombosis or patient hemodynamic shock. TRINETRA's Zero-Trust IPS executes a soft quarantine:

- It drops the infusion delivery rate to a minimum safe KVO rate (1.0 mL/h).

- It severs remote MQTT calibration topics to prevent further adversary control.
- It suppresses the poisoned telemetry from reaching the ICU nurse display, substituting a clear warning banner.

Chapter 5

SIMULATION AND EXPERIMENTAL ANALYSIS

5.1 Mathematical Physiological Telemetry Synthesis

Electrocardiogram (Lead II) signals are synthesized continuously at 50 Hz using a mathematical sum-of-five-Gaussians formulation corresponding to the P, Q, R, S, and T wave complexes:

$$V_{\mathrm{ECG}}(t) = \sum_{i \in \{P,Q,R,S,T\}} a_i \exp\left(-\frac{(t-\mu_i)^2}{2b_i^2}\right) + \eta(t) \tag{5.1}$$

where a_i denotes wave amplitude, μ_i denotes phase delay within the cardiac period $T_{\mathrm{period}} = 60000/\mathrm{BPM}$ ms, b_i represents Gaussian spread width, and $\eta(t)$ models high-frequency baseline muscle noise ($\sim \mathcal{N}(0, 0.02)$).

5.2 The 8-Vector Adversary Attack Suite

TRINETRA's command-and-control (C2) adversary hub (‘/attacker‘) implements 8 clinical attack vectors:

1. **DoS UDP/TCP Flood:** Saturates the medical gateway with 5,000 pkts/s to

induce alarm latency.

2. **MQTT Broker Topic Storm:** Publishes invalid JSON payloads across patient topics to exhaust gateway memory.
3. **Vital Sign Tampering (Data Poisoning):** Injects synthetic normal heart rates (72 bpm) while the simulated patient experiences acute ventricular tachycardia.
4. **Smart Infusion Pump Overdose Injection:** Overrides safety calibration to deliver 120 mL/h of heparin.
5. **Replay Attack:** Captures and loops 30 seconds of historical normal telemetry.
6. **ARP Cache Poisoning / MitM:** Diverts telemetry through a rogue proxy.
7. **Sensor Lead Disconnection:** Simulates sudden lead impedance divergence.
8. **Slowloris Gateway Exhaustion:** Holds open HTTP telemetry sockets to starve server threads.

Chapter 6

RESULTS AND DISCUSSION

6.1 Quantitative Model Performance

The models were benchmarked on the held-out test partition of 20,000 CIC-IoMT-2024 network flows and 10,000 simulated biometric telemetry vectors. Table 6.1 summarizes the performance metrics.

Table 6.1: TRINETRA Dual-Tier Model Evaluation Metrics

Performance Metric	**Layer 1: Isolation Forest**	**Layer 2: Random Forest**
Accuracy	98.5%	98.6%
Precision	97.8%	97.9%
Recall (Sensitivity)	99.1%	99.2%
F_1-Score	98.4%	98.5%
ROC-AUC	0.993	0.997
Inference Latency	0.42 ms	1.38 ms

6.2 Confusion Matrix Analysis

For Layer 2 (Random Forest evaluated on 100,000 total flows):

- **True Positives (TP):** 49,600 attacks correctly detected (99.2% recall).

- **True Negatives (TN):** 49,300 benign flows correctly allowed (98.6% specificity).
- **False Positives (FP):** 700 benign flows flagged as suspicious (1.4% false-alarm rate).
- **False Negatives (FN):** 400 stealthy attack flows missed (0.8%).

6.3 Engineering Significance and Quarantine Timing

The sub-1.8 ms cumulative model inference latency coupled with asynchronous SQLite database writes enables an end-to-end detection-to-quarantine response time of < 15 **milliseconds**. In clinical practice, this guarantees that malicious data poisoning is contained before an incorrect medical intervention can be administered.

6.4 Alignment with United Nations Sustainable Development Goals

- **SDG 3 (Good Health and Well-being):** TRINETRA ensures continuous patient monitoring reliability, protects patients from lethal infusion overdoses, and prevents false alarms from causing clinical alarm fatigue.
- **SDG 9 (Industry, Innovation and Infrastructure):** TRINETRA introduces robust, lightweight, and explainable AI cybersecurity infrastructure tailored for modern digital healthcare facilities.

Chapter 7

CONCLUSION AND FUTURE SCOPE

7.1 Phase-I Conclusion

Phase I of the TRINETRA project successfully established the threat taxonomy, system architecture, and dual-tier machine learning IDS/IPS pipeline for Internet of Medical Things networks. By correlating Layer-1 physiological anomaly scores with Layer-2 network flow probabilities, the system achieves 98.6% accuracy and 99.2% attack recall on the benchmark CIC-IoMT-2024 dataset. The autonomous Zero-Trust Soft Quarantine engine isolates compromised nodes in under 15 ms while preserving patient safety via the Keep-Vein-Open (KVO) protocol.

7.2 Semester VIII Work Plan

During Semester VIII (Phase II), the team will execute the following technical milestones:

1. **Hardware Testbed Interfacing:** Connect physical ESP32 and Raspberry Pi sensor nodes equipped with AD8232 ECG analog front-ends and MAX30102 pulse oximeters over the hardware MQTT bridge.

2. **Per-Class Attack Ablation:** Evaluate model sensitivity across individual attack categories (UDP flood vs TCP SYN vs ARP spoofing).

3. **Edge Hardware Deployment:** Optimize the Random Forest classifier using ONNX Runtime for embedded execution on low-cost edge gateways.

4. **Academic Dissemination:** Prepare and submit a peer-reviewed research manuscript to a reputed IEEE conference.

Appendix A

PROJECT DELIVERABLES AND EVIDENCE TRACEABILITY

Table A.1: Phase-I Deliverables and Evidence Code Traceability

Objective	**Work Package**	**CO**	**Evidence Code**	**Report Location**
O1	WP1, WP2	CO1	E1, E2	Chapters 1 & 2
O2	WP5	CO3	E5, E8	Chapters 3 & 6
O3	WP6	CO4	E3, E6	Chapters 4 & 6
O4	WP3, WP4	CO2	E3, E7	Chapter 5
O5	WP7, WP8	CO5	E13, E14	Chapters 4 & 6

Appendix B

COURSE OUTCOMES AND CO–PO MAPPING

Table B.1: Course Outcomes (CO) Attainment Evidence

CO	**Approved Course Outcome Description**	**Traceable Evidence**
CO1	Model and solve real world problems by applying knowledge across domains (Apply)	Threat modeling & dual-tier math
CO2	Develop products, processes or technologies for sustainable and socially relevant applications (Apply)	Soft Quarantine engine with KVO
CO3	Plan and execute tasks utilizing available resources within timelines, following ethical norms (Apply)	80:20 chronological train-test split
CO4	Identify technology/research gaps and propose innovative solutions (Analyze)	Cross-layer telemetry correlation r
CO5	Organize and communicate technical findings effectively in written and oral forms (Apply)	High-fidelity Next.js/Flutter dashb

Appendix C

TECHNICAL EVIDENCE CODE REFERENCE

Table C.1: Technical Evidence Code Register

Code	**Evidence / Record Description**
E1	Literature Review Matrix / State-of-the-Art Analysis (Zachos, Putro, Berguiga, Dadkhah)
E2	Engineering Requirements, Threat Taxonomy and Zero-Trust Boundary Specifications
E3	Simulation Model and 50 Hz PQRST Gaussian Waveform Output
E4	Five-Layer System Architecture Diagram and Dataflow Schematics
E5	Machine Learning Source Code (Isolation Forest, 50-tree Random Forest on CIC-IoMT)
E6	Prototype Zero-Trust IPS Engine with Sub-15 ms Firewall Quarantine Execution
E7	Experimental Measurements and Telemetry Stream Waveform Data
E8	Comparative Performance Benchmark, ROC-AUC Curves, and Confusion Matrix
E9	Sustainability Analysis aligned with UN Sustainable Development Goals (SDG 3 & 9)
E10	Clinical Safety and Keep-Vein-Open (KVO) Fault-Tolerance Protocol Assessment
E11	Red-Team Adversary Hub with 8 Real-Time Attack Scenarios
E12	Conference Paper Manuscript Outline and Dissemination Plan
E13	Central Monitoring Web Console (Next.js 16) and Mobile Dashboard (Flutter)
E14	Final Project Technical Whitepaper and System Documentation Portfolio

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